

Unit ~ 1

__/__/

~ Relations

① Reflexive

R is a relation on set A

R is reflexive if $(a,a) \in R \quad \forall a \in A$

② Symmetric

if $(a,b) \in R \Leftrightarrow (b,a) \in R$

③ Irreflexive

if $(a,a) \notin R \quad \forall a \in A$

④ Anti symmetric

if $(a,b) \in R \ \& \ (b,a) \in R \Leftrightarrow$

⑤ Asymmetric

if $(a,b) \in R \Leftrightarrow (b,a) \notin R$

⑥ Transitive

if $(a,b) \in R \ \& \ (b,c) \in R \Leftrightarrow (a,c) \in R$

Relation	Ref	Irref ($A = \{1,2\}$)	Sym	Asym	Antisym	Trans
$R_1 = \{(1,1), (2,2)\}$	✓	X	✓	X	✓	✓
$R_2 = \{(1,2), (2,1)\}$	X	✓	✓	X	X	X
$R_3 = \{(1,2)\}$	X	✓	X	✓	✓	✓
$R_4 = \{(1,1), (1,2)\}$	X	✓	X	✓	✓	✓
$R_5 = \{(1,1), (2,2), (1,2)\}$	✓	X	X	✓	✓	✓
$R_6 = \{(1,1), (2,2), (1,2), (2,1)\}$	✓	X	✓	X	X	✓

⑦ Equivalence Relation

Any relation R on set A is said to be an equivalence relation if R is reflexive, symmetric & transitive.

⑧ Partial order relation

Any relation R on set A is said to be a partial order relation if it's reflexive, antisymmetric & transitive.

⑧ R_5 is a relation on set of integers such that any ordered pair belongs to R if $x \equiv y \pmod{5}$
 $R_5 = \{(x, y) \mid x \equiv y \pmod{5}\}$

$x - y$ is divisible by 5

Let's take (a, a)

$$a - a = 0$$

0 is divisible by 5

$$\Rightarrow (a, a) \in R_5 \quad \forall a \in \mathbb{Z}$$

R_5 is reflexive

Let's suppose $(a, b) \in R_5$

$$\Rightarrow a - b = 5K$$

$$\Rightarrow b - a = 5X$$

$$\Leftrightarrow (b, a) \in R_5$$

R_5 is symmetric

Let's suppose $(a, b) \in R_5$ & $(b, c) \in R_5$

$$\Rightarrow a - b = 5K \quad \Delta \quad b - c = 5Z$$

add

$$\Rightarrow a - c = 5P$$

$$\Leftrightarrow (a, c) \in R_5$$

R_5 is transitive

Hence, R_5 is equivalence relation on $\mathbb{Z}$

For Partial order

Since

$$(1, 6) \in R \quad \& \quad (6, 1) \in R$$

$$\text{but } 1 \neq 6$$

Hence, it's not anti sym
so it's not partial order relation

~ Closure of a relation

R is a relation on set A

① Reflexive closure

$$\Delta A = \{(a, a) \mid a \in A\}$$

↓

Diagonal Relation

$R \cup \Delta A \rightarrow$ Reflexive closure of R

↓

R me kya add krni ki reflexive hojaye

② Symmetric closure

$$R \cup R^{-1}$$

③ Transitive closure

$$R^* = R \cup R^2 \cup R^3 \dots \cup R^n$$

$n \rightarrow$ no of elements in A

$$R^n = R^{n-1} \circ R$$

composition

or represent in matrix form to find R^n

$$Q \quad A = \{1, 2, 3, 4\}$$

$$R = \{(1,1) (1,2) (2,1) (3,2) (2,4)\}$$

Matrix form

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

(1,1) hai to $a_{11} = 1$ but (3,1) mi hai to $a_{31} = 0$ ese hi puri matrix banalo

$$R^2 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R^3 = \begin{bmatrix} 2 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 0 & 1 \\ 2 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R^4 = \begin{bmatrix} 3 & 2 & 0 & 1 \\ 2 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 3 & 0 & 2 \\ 3 & 2 & 0 & 1 \\ 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now,

$$R_1 \cup R_2 \cup R_3 \cup R_4 = A \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

ek bar bi agr $a_{ij} \geq 1$ hai to 1 else 0

$$R^* = \{(1,1), (1,2), (1,4), (2,1), (2,2), (2,4), (3,1), (3,2), (3,4)\}$$

→ Equivalence Classes

Equivalence class of an elementary set A consists of all elements related to a by an equivalence relation

$$[a] = \{b \mid aRb\} \quad R \rightarrow \text{equivalence relation}$$

$$A = \{1, 2, 3\}$$

$$R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$$

$$[1] = \{1, 2\}$$

$$[2] = \{1, 2\}$$

$$[3] = \{3\}$$

→ Partition of a set
dividing a set A into subsets $A_1, A_2, \dots, A_n$ such that

$$\bigcup_{i=1}^n A_i = A$$

$$\bigcap_{i=1}^n A_i = \phi \quad (A_i \cap A_j = \phi) \quad \begin{matrix} i=[1, n] \\ j=[1, n] \end{matrix}$$

$$A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \\ A_1 = \{1, 2, 3\} \quad A_2 = \{6\} \quad A_3 = \{4, 5\} \quad A_4 = \{7, 8, 10\}$$

Equivalence class divides a set into partition

$$A = \{1, 2, 3, 4\}$$

$$R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4), (4,3)\}$$

$$[1] = \{1, 2\}$$

$$[2] = \{1, 2\}$$

$$[3] = \{3, 4\}$$

$$[4] = \{3, 4\}$$

divided into 2 partitions

→ Theorem

Let R is an equivalence relation on set A then

- $a \in [a] \quad \forall a \in A$
- $[a] = [b] \quad \text{if } aRb$
- either $[a] = [b]$ or $[a]$ & $[b]$ are disjoint

• Proof

$$i) (a, a) \in R \quad \forall a \in A$$

[$\because R$ is reflexive]

$$\text{if } (a, a) \in R \Leftrightarrow a \in [a] \quad \forall a \in A$$

ii) Let $x \in [a]$

$\Rightarrow (x, a) \in R$

Also, $aRb \Rightarrow (a, b) \in R$

$(x, a) \& (a, b) \in R$

$\Leftrightarrow (x, b) \in R$ (Since R is transitive)

$\Rightarrow x \in [b]$

$\Rightarrow [a] \subseteq [b]$

Let $y \in [b]$

$\Rightarrow (y, b) \in R$

also $bRa \Rightarrow (b, a) \in R$

$(y, b) \& (b, a) \in R$

$\Leftrightarrow (y, a) \in R$ (Since $R \rightarrow$ transitive)

$\Rightarrow y \in [a]$

$\Rightarrow [b] \subseteq [a]$

As $[a] \subseteq [b] \& [b] \subseteq [a] \Leftrightarrow [a] = [b]$

iii) Let $x \in [a] \& x \in [b]$

$\Rightarrow (x, a) \in R \& (x, b) \in R$

$\Rightarrow (a, x) \in R \& (x, b) \in R$ ($R \rightarrow$ symmetric)

$\Rightarrow (a, b) \in R$ ($R \rightarrow$ transitive)

$\Rightarrow aRb$

from (ii) as $aRb \Leftrightarrow [a] = [b]$

$\rightarrow$ Composition of relations is associative

Let R, S, T are 3 relations

then $(R \circ S) \circ T = R \circ (S \circ T)$

Let $(a, d) \in (R \circ S) \circ T$

$\Rightarrow \exists c$ s.t. $(a, c) \in R \circ S \& (c, d) \in T$

$\Rightarrow \exists b$ s.t. $(a, b) \in R \& (b, c) \in S \& (c, d) \in T$

$\Rightarrow (a, b) \in R \& (b, d) \in \overline{S \circ T}$

$\Leftrightarrow (a, d) \in R \circ (S \circ T)$

Hence

$$(R \circ S) \circ T \subset R \circ (S \circ T)$$

$$\text{//ly } R \circ (S \circ T) \subset (R \circ S) \circ T$$

$$\Leftrightarrow (R \circ S) \circ T = R \circ (S \circ T)$$

Hence, composition of relations is associative

Q P.T $A \times (B \cap C) = (A \times B) \cap (A \times C)$

Let $(a, b) \in A \times (B \cap C)$

$$\Rightarrow a \in A \text{ and } b \in B \cap C$$

$$\Rightarrow b \in B \text{ and } b \in C$$

$$\text{Since } a \in A \text{ \& } b \in B \Leftrightarrow (a, b) \in A \times B$$

$$\text{Since } a \in A \text{ \& } b \in C \Leftrightarrow (a, b) \in A \times C$$

$$\text{As } (a, b) \in A \times B \text{ \& } (a, b) \in A \times C$$

$$\Leftrightarrow (a, b) \in (A \times B) \cap (A \times C)$$

$$\Rightarrow A \times (B \cap C) \subset (A \times B) \cap (A \times C)$$

$$\text{//ly, } (A \times B) \cap (A \times C) \subset A \times (B \cap C)$$

$$\Rightarrow A \times (B \cap C) = (A \times B) \cap (A \times C)$$

Q Show that for set A

$$A \times \phi = \phi \times A = \phi$$

Let $(a, b) \in A \times \phi$

$\Rightarrow a \in A$ & $b \in \phi$

Since $b \in \phi \Rightarrow b = \emptyset$

Since $b \notin \phi$ ($\phi \rightarrow$ empty)

$\therefore$ No pairing can be done

$$\Rightarrow A \times \phi = \phi$$

$$\text{Similarly } \phi \times A = \phi$$

Q Suppose that $A \times B = \phi$

$\Rightarrow$ either A or B or both A & B are null sets

Q Let A is a set of non zero integers. Let this is a relation defined on $A \times A$ defined by $(a, b) \approx (c, d)$ if $ad = bc$

P.T $\approx$ is equivalence

Let $(a, b) \in A \times A$

$$\Rightarrow (a, b) \approx (a, b) \quad \because \quad a \cdot b = a \cdot b \\ b \cdot a = b \cdot a$$

$\Rightarrow \approx$ is reflexive

Let $(a, b) \approx (c, d)$

Let $(c, d) \approx (e, f)$

$$\Rightarrow ad = bc$$

$$ad = bc \text{ \& } cf = de$$

$$\Rightarrow bc = ad$$

$$d/c = b/a \quad d/c = f/e$$

$$\Rightarrow cb = da$$

$$\Rightarrow b/a = f/e$$

$$\Rightarrow (c, d) \approx (a, b)$$

$$\Rightarrow be = af$$

$\Rightarrow \approx$ is sym

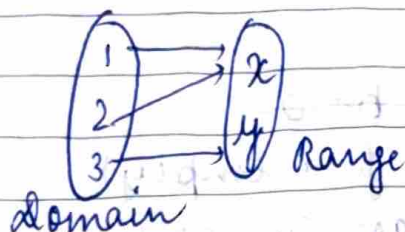
$$\Rightarrow (a, b) \approx (e, f)$$

$\Rightarrow \approx$ is transitive

$\Rightarrow \approx$ is equivalence relation

→ Functions

To each element of set A assign a unique element of set B $f: A \rightarrow B$ $A = \{1, 2, 3\}$ $B = \{x, y\}$



→ Types of functions

• One-One

Each element has diff image $x \neq y \Leftrightarrow f(x) \neq f(y)$

• Many-One

Two or more elements have same image

• Onto

Every element of B is img of some element of A

• Into

Atleast one element of B isn't img of any element of A

→ Inverse of a function (one-one & onto hona chahiye)

$$f: A \rightarrow B$$

$$f^{-1}: B \rightarrow A$$

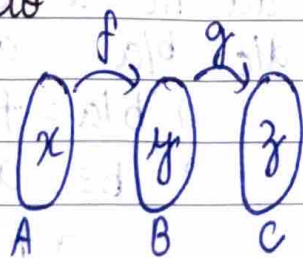
→ Composition of a function

$$f: A \rightarrow B$$

$$g: B \rightarrow C$$

$$g \circ f: A \rightarrow C$$

Q $x \neq y$ $f(x) = f(y)$ $f \& g$ are onto prove $g \circ f$ is onto



Let $z \in C$

Since g is onto

$\exists y \in B$ st z is img of y

Since f is onto

$\exists x \in A$ st y is img of x

therefore $g \circ f$ is onto

Q $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x - 3$ f^{-1} exists?

$$f(x) = f(y)$$

$$2x - 3 = 2y - 3$$

$$\Rightarrow x = y \quad f \rightarrow \text{one one}$$

$$2x - 3 \in \mathbb{R} \quad \Rightarrow f \rightarrow \text{onto}$$

$$\Rightarrow f^{-1} \text{ exists}$$

show

Q $f(x) = x^4$, $g(x) = x^{1/4}$ $x \in \mathbb{R}$ are inverse of each other

If Two func are inverse of each other then
 $f \circ g = \text{Identity func} = g \circ f$

$$f \circ g(x) = (x^{1/4})^4 = x$$

$$g \circ f(x) = (x^4)^{1/4} = x$$

Hence f & g are inverse of each other

Q $f(x) = x + 9$ $g(x) = x^2 + 3$

$$f \circ f(a) = a + 18$$

$$g \circ g(a) = a^4 + 6a^2 + 12$$

$$f \circ g(b) = b^2 + 12$$

$$g \circ f(b) = b^2 + 18b + 84$$

$$g \circ f(4) = 172$$

$$f \circ g(-4) = 28$$

→ Propositional calculus

Proposition → Any statement that's true / false

$p: 2+2=4$ → True proposition

$q: \text{Today is Monday}$ → False proposition

Close the door → Not a proposition

p & q are propositional variables

→ Connectives

Used to combine propositional variables

→ Truth Table

Tabular form of representing a compound proposition

→ Conjunction ($\wedge$) → Both input ^{true} same to output
(AND)

→ Disjunction ($\vee$) → At least one input true to output true
(OR)

→ Negation ($\neg$) → T Ka F, F Ka T

→ Implication / Conditional ($\rightarrow$)
 $P \rightarrow Q$ if P true then Q is true

→ Biconditional ($\leftrightarrow$)

$P \leftrightarrow Q$ P if & only if Q

P	Q	$P \wedge Q$	$P \vee Q$	$\sim P$	$P \rightarrow Q$	$P \leftrightarrow Q$	$\sim P \vee Q$
T	T	T	T	F	T	T	T
T	F	F	T	F	F	F	F
F	T	F	T	T	T	F	T
F	F	F	F	T	T	T	T

★ . $(P \rightarrow Q) \cong (\sim P \vee Q)$

$\cong$ equivalent to each other

• $(P \leftrightarrow Q) \cong (P \rightarrow Q) \wedge (Q \rightarrow P)$

→ Negation of compound proposition

$\sim(P \vee Q) \approx \sim P \wedge \sim Q$

$\sim(P \wedge Q) \approx \sim P \vee \sim Q$

$\sim(\sim P \wedge Q) \approx P \vee \sim Q$

$\sim(\sim P \vee \sim Q) \approx P \wedge Q$

Q Truth Table for $(P \wedge Q) \vee (\sim Q \wedge R) = A$

P	Q	R	$\sim Q$	$P \wedge Q$	$\sim Q \wedge R$	A
T	T	T	F	T	F	T
T	T	F	F	T	F	T
T	F	T	T	F	T	T
T	F	F	T	F	F	F
F	T	T	F	F	F	F
F	T	F	F	F	F	F
F	F	T	T	F	T	T
F	F	F	T	F	F	F

→ converse, Inverse & contrapositive

if $P \rightarrow Q$ then $Q \rightarrow P$ (converse)

$\sim P \rightarrow \sim Q$ (inverse)

$\sim Q \rightarrow \sim P$ (contrapositive)

$P \rightarrow F$
 $Q \rightarrow T$
 $P \rightarrow Q \rightarrow T$ } various truth

Q Prove that conditional proposition distribute over conjunction

PT

$$(P \rightarrow (Q \wedge R)) \cong (P \rightarrow Q) \wedge (P \rightarrow R)$$

P	Q	R	$P \rightarrow Q$	$P \rightarrow R$	$(P \rightarrow Q) \wedge (P \rightarrow R)$	$Q \wedge R$	$P \rightarrow (Q \wedge R)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	F	F	F	F	F
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	F	F
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

same

$$Q \sim(P \leftrightarrow Q) \cong \sim P \leftrightarrow \sim Q$$

P	Q	$\sim P$	$\sim Q$	$\sim P \leftrightarrow \sim Q$	$P \leftrightarrow Q$	$\sim(P \leftrightarrow Q)$
T	T	F	F	T	T	F
T	F	F	T	F	F	T
F	T	T	F	F	F	T
F	F	T	T	T	T	F

→ Tautology
 When compound proposition statement yields true value for all possible combinations of truth table in truth table.

→ Contradiction
 False for all possible combinations

P	$\sim P$	$P \vee \sim P$	$P \wedge \sim P$
T	F	T	F
F	T	T	F

→ Contingency
 Combination of T & F values

→ $\sim(P \wedge Q) \Leftrightarrow (\sim P \vee \sim Q)$

P	Q	$P \wedge Q$	$\sim(P \wedge Q)$	$\sim P$	$\sim Q$	$\sim P \vee \sim Q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

← same →

→ $P \rightarrow (Q \wedge R) \Leftrightarrow \sim(P \rightarrow Q)$

P	Q	R	$P \rightarrow Q$	$\sim(P \rightarrow Q)$	$Q \wedge R$	$P \rightarrow (Q \wedge R)$
T	T	T	T	F	T	T
T	T	F	T	F	F	F
T	F	T	F	T	F	F
T	F	F	F	T	F	F
F	T	T	T	F	T	T
F	T	F	T	F	F	T
F	F	T	T	F	F	T
F	F	F	T	F	F	T

→ Argument

It's an assertion that a set of proposition (premises) gives a conclusion

$$[P_1, P_2, \dots, P_n : Q] \quad P_1 \wedge P_2 \wedge P_3 \dots P_n \rightarrow Q \cong T_1$$

valid argument

Q

$P_1 : P \rightarrow Q$

$P_2 : P \quad \therefore Q$

P	Q	$P \rightarrow Q$	$(P \rightarrow Q) \wedge P$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	F

Q

$P \rightarrow Q \quad Q \rightarrow R$

$\therefore P \rightarrow R$

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$(P \rightarrow Q) \wedge (Q \rightarrow R)$	$P \rightarrow R$
T	T	T	T	T	T	T
T	F	F	T	F	F	F
T	F	T	F	T	F	T
T	F	F	F	T	F	F
F	T	T	T	T	T	T
F	T	F	T	F	F	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

Q If I enjoy studying then I will study, I will do my homework or I will not study, I won't do my hw $\therefore$ I don't enjoy studying

$P \rightarrow$ I enjoy studying $P_1: P \rightarrow Q$
 $Q \rightarrow$ I will study $P_2: R \vee \sim Q$
 $R \rightarrow$ I will do my hw $P_3: \sim R$
 $\therefore \sim P$

	P_1				P_2		P_3	
P_1, P_2, P_3	P	Q	R	$P \rightarrow Q$	$\sim Q$	$R \vee \sim Q$	$\sim R$	$\sim P$
F	T	T	T	T	F	T	F	F
F	T	T	F	T	F	F	T	F
F	T	F	T	F	T	T	F	F
F	T	F	F	F	T	T	T	F
F	F	T	T	T	F	T	F	T
F	F	T	F	T	F	F	T	T
F	F	F	T	T	T	T	F	T
T	F	F	F	T	T	T	T	T

$\rightarrow$ Valid Argument Form

① Modus Ponens
 $P \rightarrow Q$
 $P \therefore Q$

② Modus Tollens
 $P \rightarrow Q$
 $\sim Q \therefore \sim P$

③ Hypothetical syllogism
 $P \rightarrow Q \quad Q \rightarrow R$
 $\therefore P \rightarrow R$

— / — / —

4. disjunctive syllogism

$$P \vee Q$$
$$\sim P \therefore Q$$

5. Constructive dilemma

$$(P \rightarrow Q) \wedge (R \rightarrow S)$$

$$P \vee R \quad \therefore Q \vee S$$

6. Absorption

$$\therefore P \rightarrow (P \wedge Q)$$

7. Simplification

$$P \wedge Q$$

$$\therefore P$$

8. Conjunction

$$\therefore P \wedge Q$$

9. Addition

$$P \vee Q$$

Q	$P \rightarrow Q$	$\left. \begin{array}{l} P \rightarrow R \text{ (3)} \\ P \rightarrow S \text{ (3)} \\ \sim P \text{ (2)} \end{array} \right\} \text{Subst (3)}$
	$Q \rightarrow R$	
	$R \rightarrow S$	
	$\sim S$	
	$P \vee T$	$\left. \begin{array}{l} \text{qu} \\ \text{su} \end{array} \right\} T \text{ (2)}$

∴ Total no. of iterations =

Q $P \rightarrow Q$
 $P \wedge R$ simplification P $\therefore Q$ (modus Ponens)

$\therefore Q$

cons. dilemma

Q $(P \rightarrow Q) \wedge (R \rightarrow S)$
 $(P \vee R) \wedge (Q \vee S)$ simplify $P \vee R$ $\rightarrow Q \vee S$

$\therefore Q \vee S$

Q $(P \wedge Q) \vee (R \rightarrow S)$ ④
 $T \rightarrow R$
 $\sim (P \wedge Q)$ $\rightarrow R \rightarrow S$
disjunct Syll $\therefore T \rightarrow S$

Hypo. Syll

$\therefore T \rightarrow S$

Q $(P \rightarrow Q) \wedge (R \rightarrow S)$
 $Q \rightarrow S$
 $(Q \rightarrow S) \rightarrow (R \vee P)$ Modus Ponenz $P \vee R$ $\rightarrow Q \vee S$

const. dilemma

$\therefore Q \vee S$

Q either I will pass the exam or I will not graduate. If I don't graduate I will go to Canada. I fail therefore I go to Canada

Find validity

$P \rightarrow$ Pass exam $Q \rightarrow$ I will graduate

$R \rightarrow$ I will go to Canada

$P \vee \sim Q$
 $\sim Q \rightarrow R$
 $\sim P$ $\therefore R$

→ Predicate

It's a word in a sentence which tells about the nature of subject

→ every rational no. is a real no.

→ Open Statement

Any declarative sentence which contains one or more variables & isn't a statement but becomes a statement when the variables are replaced by available options.

→ $p(x) : x$ is less than 10

→ Quantifiers

- Existential Quantifiers

$$\exists x p(x)$$

$p(x) : x$ is less than 10

$$\exists x \sim p(x)$$

- Universal Quantifiers

$$\forall x p(x)$$

→ Negation of Quantifiers

$$\sim (\exists x p(x)) = \forall x \sim p(x)$$

$$\sim (\forall x p(x)) = \exists x \sim p(x)$$

$$\sim (\forall x \sim p(x)) = \exists x p(x)$$

Q $p(x) : x > 0$ $r(x) : x$ is perfect sq
 $q(x) : x$ is odd $t(x) : x$ is divisible by 2

- i) Atleast one integer is odd
 $\exists x q(x)$
- ii) No odd integer is divisible by 2
 $\forall x [q(x) \rightarrow \sim t(x)]$
- iii) There exists an odd integer divisible by 2
 $\exists x [q(x) \wedge t(x)]$
- iv) If x is odd & is perfect sq & is div by 2
 $\exists x [q(x) \wedge r(x) \rightarrow t(x)]$

→ Property of Quantifiers

- $\exists x (p(x) \rightarrow q(x)) \equiv \forall x p(x) \rightarrow \exists x q(x)$
- $\exists x p(x) \rightarrow \forall x q(x) \equiv \forall x (p(x) \rightarrow q(x))$
- $\exists x (p(x) \vee q(x)) \equiv \exists x p(x) \vee \exists x q(x)$
- $\exists x (p(x) \wedge q(x)) \equiv \exists x p(x) \wedge \exists x q(x)$
- $\sim (\exists x \sim p(x)) \equiv \forall x p(x)$

Q Find negation

$$\forall x \exists y [(p(x, y) \wedge q(x, y)) \rightarrow r(x, y)]$$

$$\exists x \forall y \sim [\sim (p_{xy} \wedge q_{xy}) \vee r_{xy}]$$

$$\exists x \forall y [p_{xy} \wedge q_{xy} \wedge \sim r_{xy}]$$

Q All integers are greater than 0

$p(x) : x$ is an integer

$q(x) : x > 0$

St $\equiv \forall x (p(x) \rightarrow q(x))$

$\sim [\forall x [p(x) \rightarrow q(x)]]$

$\exists x [\sim p(x) \vee q(x)]$

$\exists x [p(x) \wedge \sim q(x)]$

→ Free & Bound Variables

→ Bound variables

If occurrence of variable is within the scope of a quantative quantifier

$\forall x p(x) \rightarrow x$ is bound

→ Free Variable

If occurrence is outside the scope of a quantifier

$\exists y \forall x p(x) \rightarrow y$ is bound & y is free

→ Closed Sentence

Which has no free variables

$\exists x (p(x) \rightarrow q(x))$

→ Disjunctive Normal Form
 A conjunction of variables / their negation is called fundamental conjunction
 DNF is disjunction of fundamental conjunction
 $(p \wedge q) \vee (\neg p \wedge q) \leftarrow \text{DNF}$

→ Conjunctive Normal Form
 Conjunction of fundamental disjunction
 $(p \vee q) \wedge (p \vee \neg q)$

Q Convert CNF $\rightarrow$ DNF & vice versa

i) $(p \rightarrow q) \wedge (\neg p \wedge q)$
 $(\neg p \wedge q) \vee (\neg p \wedge q)$

ii) $(\neg p \vee q) \wedge (\neg p \wedge q)$
 $(\neg p \wedge \neg p \wedge q) \vee (q \wedge \neg p \wedge q)$
 $(\neg p \wedge q) \vee (q \wedge \neg p \wedge q)$

iii) $\sim (P \rightarrow (Q \wedge R))$
 $\sim (\neg P \vee (Q \wedge R))$
 $\sim (\neg P \vee Q \wedge R)$
 $P \vee \sim Q \vee \sim R$
 $(P \wedge \sim Q) \vee (P \wedge \sim R)$

iv) $(P \wedge Q) \vee (\neg P \wedge Q \wedge R)$
 $(P \vee \neg P \wedge Q \wedge R) \wedge (Q \vee \neg P \wedge Q \wedge R)$
 $(P \vee \neg P) \wedge (P \vee Q) \wedge (P \vee R) \wedge (Q \vee \neg P) \wedge (Q \vee Q) \wedge (Q \vee R)$
 $(P \vee Q) \wedge (P \vee R) \wedge (Q \vee \neg P) \wedge Q \wedge (Q \vee R)$

